Causal Inference

Midterm Review Session

DS-UA 201 · October 2026

Probability, statistics, experiments, regression and natural experiments

Regression recap: two ways to read a line

Descriptive

$Y = \beta_0 + \beta_1 X + \varepsilon$

β_1 is the slope of the best linear predictor: the line that minimizes $E[(Y - \beta_0 - \beta_1 X)^2]$.

By construction $E[\varepsilon] = 0$ and $E[X\varepsilon] = 0$, which gives $\beta_1 = \text{Cov}(X, Y) / \text{Var}(X)$.

Always identified. It answers: how does Y move with X on average?

Causal

$Y(S, U) = \alpha_0 + \alpha_1 S + U$

α_1 is the effect on Y of raising S by one, holding everything else (U) fixed.

The regression slope is $\beta_1 = \alpha_1 + \text{Cov}(S, U) / \text{Var}(S)$.

Identified only if $S \perp U$. Then $\text{Cov}(S, U) = 0$ and $\beta_1 = \alpha_1$.

Same regression, same number. Whether β_1 is causal depends on the assumption. Ice cream sales predict crime ($\beta_1 > 0$), but the causal effect is zero: temperature sits in U.

Binary treatment: regression is a difference in means

When S is 0 or 1, the regression slope simplifies (we derive this in Problem 8):

$$\beta_1 = \mathrm{Cov}(S, Y) / \mathrm{Var}(S) = E[Y \mid S = 1] - E[Y \mid S = 0]$$

If $S \perp U$, as in a good experiment or natural experiment, that difference in means is the **ATE**. So regressing Y on a treatment dummy recovers the ATE.

Estimation. The OLS estimator $\hat{\beta}_1$ = sample Cov(S, Y) / sample Var(S) is consistent. With binary S it equals the difference in sample means exactly.

Why use regression at all? It gives standard errors, p-values and confidence intervals, and lets us pick the right type of standard error.

Standard errors: robust vs. clustered

	Robust (HC0)	Clustered
Use when	Large iid sample; treatment assigned person by person	Treatment assigned, or sample drawn, in groups
Assumes	Observations are independent	Clusters are independent, and there are many of them
Acts like a sample of	n individuals	The number of clusters
Microfinance result	SE 193, p = 0.050	SE 199, p = 0.058

The point estimate is identical (377.79) either way. Only the uncertainty changes.

People in the same neighborhood share shocks, so they are not independent pieces of information. Robust SEs ignore this and make us overconfident. **Rule: cluster at the level treatment was assigned** (or the level groups were sampled).

Reading regression output

Microfinance, investment on treatment, clustered by neighborhood:

Output	Value	What it means
coef (treatment)	377.79	β_1: treated minus control mean investment, in rupees
std err	199.15	Uncertainty in the estimate (clustered)
z	1.897	coef / SE: how many standard errors the estimate is from 0
P>|z|	0.058	Chance of a z at least this extreme if the true effect were 0
[0.025, 0.975]	[−12.5, 768.1]	95% CI ≈ coef ± 1.96 × SE; it includes 0
Intercept	280.07	Average investment in control neighborhoods, $E[Y \mid S = 0]$
R^2	0.001	Treatment explains little of the variation; irrelevant to whether β_1 is causal

> At the 5% level we fail to reject 0 ($p = 0.058$, and the CI contains 0). With robust SEs, $p = 0.050$: right on the line. The type of SE can change the conclusion.

More regression facts worth knowing

- **The line passes through the means:** $\beta_0 = E[Y] - \beta_1 E[X]$.
- **Residual facts are built in:** $E[\varepsilon] = 0$ and $Cov(X, \varepsilon) = 0$ hold by construction. They are not assumptions, so they say nothing about $S \perp U$.
- **Omitted variable bias:** if $Y = \alpha_0 + \alpha_1 S + U$, then $\beta_1 = \alpha_1 + Cov(S, U) / Var(S)$. The sign of $Cov(S, U)$ gives the direction of the bias.
- **Best linear predictor:** if the relationship is curved, transform X (X^2, log X). It is still linear regression.
- **R^2 measures fit, not causality.**
- **Binary S:** OLS gives exactly the difference in means, the same estimate as a two-sample comparison.
- **Microfinance was cluster-randomized** (whole neighborhoods) because loans spill over to neighbors and a lender can't serve only some households. The estimate combines direct and spillover effects.

From experiments to observational data

Experiments work because $S \perp U$. That assumption has a **reason** (we randomized) and a **consequence** (U has the same distribution in treatment and control). When experiments are too costly, slow or unethical, we try to recover one of the two.

Mimic the reason

Find randomness that happened "naturally" in the world, then argue $S \perp U$ because treatment was as-if random.

Natural experiments (today), sometimes with controls or instruments.

Mimic the consequence

Find or build a control group that looks like the treated group, so $f(u \mid S = 1) = f(u \mid S = 0)$.

Matching, regression discontinuity, difference-in-differences (later).

Natural experiments: treat it like an experiment

A natural experiment is when we argue that some real-world process assigned treatment as-if randomly. If $S \perp U$ holds, identification and estimation work exactly as in an experiment. The key is the context: **why** was treatment essentially random?

Assumption	Question to ask	If it fails
$S \perp U$	Why was treatment random? Random among whom?	Narrow the population to where it is random
Perfect compliance	Does everyone follow the random process?	Estimate the ITT instead
SUTVA	Can one unit's treatment affect another's outcome?	Define treatment at the group level; cluster SEs
Balance test	Do observables look the same across groups?	Unobservables need institutional context

Natural experiments: from assumption to estimate

1. Assumption. $S \perp U$ within a defined population (e.g. lottery players).

2. Identification. Same argument as an experiment: $ATE = E[Y(1, U) \mid S = 1] - E[Y(0, U) \mid S = 0] = E[Y \mid S = 1] - E[Y \mid S = 0]$.

3. Estimation. Difference in sample means, or regress Y on S. The slope is the same number.

4. Inference. Cluster SEs at the level where the randomness happened (the postcode).

5. Interpretation. Say exactly what was randomized. In the lottery: the effect of living in a winning postcode, among players, including the effect of neighbors winning too.

Compliance here means following the real-world random process, not an experimenter. **SUTVA fails** but treatment is random by group? Analyze at the group level. **Compliance fails?** Estimate the ITT.

Worked example: the Dutch Postcode Lottery

1. Question. How does a sudden €12,500 change spending? Giving money away at random is too expensive, but the lottery does it for us (Kuhn et al., 2011).

2. Population. You can only win if you play, and choosing to play is not random. So $S \perp U$ is only plausible *among players*: compare winners to non-winners who played.

3. Treatment. A whole postcode wins together, so neighbors may affect each other (SUTVA). Treatment is defined as living in a winning postcode: a natural cluster-randomized experiment.

4. Compliance. Plausible: nobody turns down their winnings.

5. Balance. Most pre-lottery traits match. A few starred rows are expected: 5% of true nulls are rejected at the 5% level. A rigged lottery would not show up here, so we lean on the institutional context.

6. Result. Regress durable spending on win_code, clustering by postcode: +€627 six months later (SE 295, $p = 0.034$).

Reading the lottery balance table

Kuhn et al., Table 1. Compare **players only**: non-winning participants (column 1) vs. winning participants (column 3).

Pre-lottery characteristic	Non-winners	Winners	Significant?
Persons in household	2.74	2.73	No
Age	49.60	52.49	Yes (**)
Higher vocational or university	0.326	0.256	Yes (*)
Food away from home (€/month)	83.85	119.63	Yes (***)
Total annual household income (€)	25,341	27,322	No
Sample size	301	223	

> Most of the roughly 25 rows match closely. A few stars are expected: at 5%, about 1 in 20 true nulls is rejected by chance. A balance test can't see unobservables, such as a rigged lottery, so credibility comes from the institutional context.

Conditional expectation: the average within a group

E[Y | X = x] is the average of Y among only the units with X = x. Conditioning just means: **restrict to a group, then average.**

Example: 10 students. S = 1 if a student was tutored, Y = test score.

Group	Students	Average score	In notation
Tutored (S = 1)	4	80	E[Y | S = 1] = 80
Not tutored (S = 0)	6	70	E[Y | S = 0] = 70
Everyone	10	74	E[Y] = 74

With no value plugged in, **E[Y | S]** is itself a random variable: it equals 80 for a tutored student and 70 for an untutored one. It depends only on S.

In words: E[Y] asks “what is the average score?” E[Y | S = 1] asks “what is the average score among tutored students?”

The LIE: the overall average is an average of group averages

$$E[Y] = E[\,E[Y \mid X]\,]$$

Read it from the inside out. **Inside:** $E[Y \mid X]$ is the average of Y in each group. **Outside:** average those group averages, weighting each group by how common it is.

With two groups (binary X), this becomes the version you will write on the exam:

$$E[Y] = E[Y \mid X = 1]\cdot P(X = 1) + E[Y \mid X = 0]\cdot P(X = 0)$$

Tutoring example: $80 \times 0.4 + 70 \times 0.6 = 32 + 42 = 74$, the same as averaging all 10 students directly.

It works for **events** too: $E[Y] = E[Y \mid A]\cdot P(A) + E[Y \mid \text{not } A]\cdot P(\text{not } A)$. On the practice exam, A was the event I $\leq 200{,}000$.

Why it's useful: when a problem hands you groups (responders, fans, treated, top-coded), the LIE splits one hard average into pieces you know something about.

Three things that show up often in proofs

Move	Rule	Why it's true	Example
1. Split by group (LIE)	$E[Y] = E[Y \mid X = 1]\cdot p + E[Y \mid X = 0]\cdot(1 - p)$	The overall average is a weighted average of group averages	$E[I] = E[I \mid R = 1]\cdot p + E[I \mid R = 0]\cdot(1 - p)$
2. Pull out what you know	$E[h(X)\cdot Y \mid X] = h(X)\cdot E[Y \mid X]$	Once you condition on X, X is a fixed number, so it moves outside like a constant	$E[XY \mid X] = X\cdot E[Y \mid X]$
3. Use independence	If $X \perp Y$, then $E[Y \mid X] = E[Y]$	Knowing X does not change the average of Y	$E[X\cdot E[Y \mid X]] = E[X\cdot E[Y]] = E[X]E[Y]$

Binary shortcuts: if S is 0 or 1, then $E[S] = P(S = 1) = p$ and $S^2 = S$. Also $SY = Y$ when $S = 1$ and $SY = 0$ when $S = 0$, so $E[SY \mid S = 1] = E[Y \mid S = 1]$ and $E[SY \mid S = 0] = 0$.

Part I. True or false (1–7)

1. For a continuous random variable X, the density f(x) tells us P(X = x), just as a PMF does for a discrete variable.

2. If Cov(X, Y) = 0, then X and Y are independent.

3. If the correlation between X and Y is 0, their covariance is also 0.

4. A p-value of 0.03 means there is a 3% probability that the null hypothesis is true.

5. We computed a 95% confidence interval of [1.2, 3.4]. There is a 95% probability that the true parameter lies in this interval.

6. Every unbiased estimator is also consistent.

7. By the CLT, the sample mean of a large iid sample is approximately normally distributed, even if the data themselves are not normal.

Part I. True or false (8–14)

8. Lowering the significance level from 5% to 1% lowers the probability of a Type II error.

9. Even if S is not independent of U, OLS still consistently estimates the descriptive slope of Y on S.

10. A regression with a high R^2 is more likely to show a causal effect.

11. Switching from robust to clustered standard errors changes the OLS estimate of the slope.

12. If a balance test finds no significant difference on any observable, we have shown that $S \perp U$.

13. In a randomized experiment with imperfect compliance, comparing average outcomes by assignment still identifies a causal effect.

14. A cluster-randomized trial identifies the ATE of the treatment on an individual.

Part I. Answers (1–4)

Statement	Answer	Why
1. The density f(x) gives P(X = x).	False	For continuous X, P(X = x) = 0. Probability is the area under f over an interval, and f(x) can even exceed 1.
2. Cov(X, Y) = 0 means X and Y are independent.	False	Zero covariance rules out only a linear relationship. If X is symmetric around 0 and $Y = X^2$, Cov = 0 but Y depends entirely on X.
3. Correlation 0 means covariance 0.	True	Corr = Cov / (SD(X)·SD(Y)), and the SDs are positive, so one is zero exactly when the other is.
4. p = 0.03 means a 3% chance the null is true.	False	The p-value is the probability of a test statistic at least as extreme as ours, assuming the null is true. It is not a probability about the null.

Part I. Answers (5–8)

Statement	Answer	Why
5. 95% chance that θ is in [1.2, 3.4].	False	The 95% describes the procedure: across repeated samples, 95% of intervals built this way contain θ. A fixed interval either contains it or not.
6. Every unbiased estimator is consistent.	False	$\hat{\theta} = X_1$ is unbiased for μ, but its variance stays σ^2 however large n gets.
7. CLT: the sample mean is approximately normal even if the data are not.	True	That is the point of the CLT: the z-score of the sample mean is approximately standard normal for large n, whatever the data's distribution (with finite variance).
8. A 1% level lowers Type II error.	False	Rejecting becomes harder, so we fail to reject false nulls more often. Fewer Type I errors, more Type II.

Part I. Answers (9–11)

Statement	Answer	Why
9. Without $S \perp U$, OLS still estimates the descriptive slope.	True	$\beta_1 = \text{Cov}(Y, S) / \text{Var}(S)$ is always identified as a descriptive parameter. $S \perp U$ is only needed to call it causal.
10. High R^2 means more likely causal.	False	R^2 measures how well the line fits, not why. Ice cream sales can predict crime well with zero causal effect.
11. Clustered SEs change the slope estimate.	False	Clustering changes only the standard errors, p-values and confidence intervals. The point estimate is identical.

Part I. Answers (12–14)

Statement	Answer	Why
12. No significant imbalance proves S ⊥ U.	False	Balance tests only check observables. Unobservables can still differ, so balance supports S ⊥ U but never proves it.
13. With imperfect compliance, comparing by assignment identifies a causal effect.	True	Assignment is still random, so the ITT, the effect of being assigned, is identified. The ATE of actually receiving treatment may not be.
14. A cluster-randomized trial identifies the individual ATE.	False	It identifies the effect of the whole group being treated, which mixes the direct effect with spillovers. The design cannot separate them.

Problem 1. Independence and covariance

Suppose $X \perp Y$, so that $E[Y \mid X] = E[Y]$.

(a) Using the Law of Iterated Expectations and the properties of conditional expectations, show that $\mathrm{Cov}(X, Y) = 0$. Justify every step except simple algebra.

(b) Does the converse hold? If $\mathrm{Cov}(X, Y) = 0$, must X and Y be independent? Give an example.

Hint: start from $\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$, then condition $E[XY]$ on X.

Problem 1. Solution (a)

(a) If $X \perp Y$, so that $E[Y \mid X] = E[Y]$, show that $Cov(X, Y) = 0$.

Step	Justification
$Cov(X, Y) = E[XY] - E[X]E[Y]$	Definition of covariance
$E[XY] = E[\, E[XY \mid X] \,]$	Law of Iterated Expectations
$= E[\, X \cdot E[Y \mid X] \,]$	Given X, X is a constant: $E[h(X)Y \mid X] = h(X)E[Y \mid X]$
$= E[\, X \cdot E[Y] \,]$ - key independence step	Independence: $E[Y \mid X] = E[Y]$
$= E[Y] \cdot E[X]$	E[Y] is a constant: $E[bX] = bE[X]$
$Cov(X, Y) = E[X]E[Y] - E[X]E[Y] = 0$	Substitute back

In words: to find E[XY], first fix X and average Y within that group (the inner expectation), then average over X (the outer one). Inside, X is just a number, so it moves out front. Independence says Y's group average is the same for every X, so E[Y | X] is simply the constant E[Y].

Problem 1. Solution (b)

(b) If Cov(X, Y) = 0, must X and Y be independent?

> No. Let X be −1, 0 or 1 with equal probability and $Y = X^2$. Then $E[X] = 0$ and $E[XY] = E[X^3] = 0$, so $Cov(X, Y) = 0$, yet knowing X tells you Y exactly.

In words: independence ⇒ Cov = 0, but not the other way around. Covariance only picks up straight-line relationships, and $Y = X^2$ is a perfect curve: the positive and negative X values cancel out.

Problem 2. Variance of a combination

(a) Starting from the definition $\mathrm{Var}(W) = E[(W - E[W])^2]$ and properties of expectations, show that

$$\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab \cdot \mathrm{Cov}(X, Y)$$

(b) Use (a) to write $\mathrm{Var}(X - Y)$.

(c) The treatment sample mean $\bar{Y}_1$ uses n_1 people with variance σ_1^2; the control mean $\bar{Y}_0$ uses no people with variance σ_0^2. The groups are independent. Find $\mathrm{Var}(\bar{Y}_1 - \bar{Y}_0)$. Why do we care about this number?

Problem 2. Solution (a)

(a) Show $Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2ab \cdot Cov(X, Y)$.

$E[(aX + bY)^2] - [E[aX+bY]]^2 = Var(aX+bY)$

$E[(a^2X^2+b^2Y^2+2abXY)] - (E[aX]+E[bY])^2$

$a^2E[X^2] + b^2E[Y^2] + 2ab\ E[XY] - (E[aX]^2 + 2E[aX]E[bY] + E[bY]^2)$

$a^2E[X^2] + b^2E[Y^2] + 2ab\ E[XY] - a^2E[X]^2 - 2ab\ E[X]E[Y] - b^2E[Y]^2$

$a^2(E[X^2]-E[X]^2) + b^2(E[Y^2]-E[Y]^2) + 2ab(E[XY]-E[X]E[Y])$

$= a^2Var(X) + b^2Var(Y) + 2ab\ Cov(X,Y)$

Justify each step: line 1 is the definition of variance; lines 2–4 use linearity, $E[X + Y] = E[X] + E[Y]$ and $E[aX] = aE[X]$; line 5 regroups; line 6 uses the definitions $Var(X) = E[X^2] - E[X]^2$ and $Cov(X, Y) = E[XY] - E[X]E[Y]$. With $a = b = 1$ this is exactly the practice exam's $Var(X + Y)$.

Problem 2. Solution (b–c)

(b) Write Var(X − Y).

> Set a = 1, b = −1: Var(X − Y) = Var(X) + Var(Y) − 2Cov(X, Y). Variances add even when we subtract.

(c) Find Var($\bar{Y}_1 - \bar{Y}_0$) for independent groups. Why do we care?

> Independent groups give Cov = 0, so Var($\bar{Y}_1 - \bar{Y}_0$) = $\sigma_1^2/n_1 + \sigma_0^2/n_0$. Its square root is the standard error of the difference in means: the denominator of the two-sample t-test.

Problem 3. Who answers the survey?

You want average household income, $E[I]$. You survey a random sample, but not everyone responds. Let $R = 1$ if a household responds and $R = 0$ if not, with $P(R = 1) = p$ and $0 < p < 1$. You compute the sample mean of income among respondents only, so its expected value is $E[I \mid R = 1]$.

(a) Using the LIE, write $E[I]$ in terms of $E[I \mid R = 1]$ and $E[I \mid R = 0]$.

(b) Show that the bias, $E[I \mid R = 1] - E[I]$, equals $(1 - p)(E[I \mid R = 1] - E[I \mid R = 0])$.

(c) Suppose higher-income households are less likely to respond. What is the sign of the bias? Under what condition is the bias zero?

Problem 3. Solution (a–b)

Setup: $R = 1$ if a household responds, $P(R = 1) = p$. The respondent mean has expected value $E[I \mid R = 1]$.

(a) Write $E[I]$ in terms of $E[I \mid R = 1]$ and $E[I \mid R = 0]$.

> R is binary, so by the LIE: $E[I] = E[I \mid R = 1] \cdot p + E[I \mid R = 0] \cdot (1 - p)$

(b) Show the bias $E[I \mid R = 1] - E[I]$ equals $(1 - p)(E[I \mid R = 1] - E[I \mid R = 0])$.

> First, why the respondent mean averages to $E[I \mid R = 1]$: $E[(1/n)\Sigma I_i] = (1/n)\Sigma E[I_i] = E[I \mid R = 1]$ (linearity; every respondent comes from the same distribution).
> $E[I \mid R = 1] - E[I] = E[I \mid R = 1] - p \cdot E[I \mid R = 1] - (1 - p) \cdot E[I \mid R = 0]$ (substitute (a))
> $= (1 - p) \cdot E[I \mid R = 1] - (1 - p) \cdot E[I \mid R = 0] = (1 - p)(E[I \mid R = 1] - E[I \mid R = 0])$

In words: the true average $E[I]$ is a mix of the two groups' averages. The respondent average misses by the share we never hear from $(1 - p)$ times how different that group is.

Problem 3. Solution (c)

Setup: bias = $(1 - p)(E[I \mid R = 1] - E[I \mid R = 0])$.

(c) Richer households respond less. Sign of the bias? When is it zero?

> $E[I \mid R = 1] < E[I \mid R = 0]$, so the bias is negative: we underestimate average income. It is zero when respondents and non-respondents have the same mean income, for example if responding is independent of income. This is selection bias: the group we observe is not the group we care about.

Compare to the practice exam (top-coding): same strategy, but split by the event $I \leq 200{,}000$ instead of R. Below the cutoff $I^T = I$, so those terms cancel; above it $I^T = 200{,}000 < E[I \mid I > 200{,}000]$, so the bias is negative.

Problem 4. Comparing estimators

$X_1, \ldots, X_n$ is an iid sample with $E[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$. Consider three estimators of μ:

$\hat{\theta}_A = X_1$ (use only the first observation)

$\hat{\theta}_B = (1 / (n + 1)) \cdot \Sigma X_i$

$\hat{\theta}_C = \bar{X}_n = (1 / n) \cdot \Sigma X_i$

(a) Find the bias of each estimator.

(b) Find the variance of each estimator.

(c) Which are consistent? Which would you use, and why?

Problem 4. Solution

Setup: iid sample, $E[X_i] = \mu$, $Var(X_i) = \sigma^2$. $\hat{\theta}_A = X_1$, $\hat{\theta}_B = \Sigma X_i / (n + 1)$, $\hat{\theta}_C = \bar{X}_n$.

(a) Bias, (b) variance and (c) consistency of each estimator.

Estimator	(a) Bias	(b) Variance	(c) Consistent?
$\hat{\theta}_A$	$\mu - \mu = 0$	σ^2	No: its variance never shrinks with n
$\hat{\theta}_B$	$n\mu/(n+1) - \mu = -\mu/(n+1)$	$n\sigma^2/(n+1)^2$	Yes: bias and variance both go to 0
$\hat{\theta}_C$	0	σ^2/n	Yes: the WLLN

> Variances use $Var(c \cdot \Sigma X_i) = c^2 \cdot \Sigma Var(X_i)$, since the covariances are 0 in an iid sample. Unbiased does not mean consistent (A), and biased does not mean useless (B is consistent, with slightly smaller variance than C). Use C: unbiased and consistent. For large n, B and C are nearly identical.

Problem 5. Naive comparisons

A city offers a voluntary financial-literacy class. Let $S = 1$ if a resident takes the class and $Y(S, U)$ be their savings one year later. Analysts compare average savings of attendees and non-attendees.

(a) Show that $E[Y \mid S = 1] - E[Y \mid S = 0] = ATT + \text{selection bias}$, where

$$ATT = E[Y(1, U) - Y(0, U) \mid S = 1]$$
$$\text{Selection bias} = E[Y(0, U) \mid S = 1] - E[Y(0, U) \mid S = 0]$$

(b) Which direction would you expect the selection bias to go here? Why?

(c) Explain why randomly assigning the class would make the selection bias zero and the ATT equal to the ATE.

Hint: add and subtract $E[Y(0, U) \mid S = 1]$.

Problem 5. Solution (a)

Setup: S = 1 if a resident chooses to take a financial-literacy class; Y = savings a year later.

(a) Show E[Y | S = 1] − E[Y | S = 0] = ATT + selection bias.

> We see Y(1, U) for attendees and Y(0, U) for everyone else, so the observed difference is
> E[Y(1, U) | S = 1] − E[Y(0, U) | S = 0]
> = E[Y(1, U) | S = 1] − E[Y(0, U) | S = 1] + E[Y(0, U) | S = 1] − E[Y(0, U) | S = 0]
> = **ATT + selection bias**

Why add and subtract? E[Y(0, U) | S = 1] is what attendees would have saved *without* the class: we never see it. Adding and subtracting it changes nothing, but splits the gap into two pieces: the class's effect on attendees (ATT), and how different attendees were to begin with (selection bias).

Problem 5. Solution (b–c)

Setup: S = 1 if a resident chooses to take a financial-literacy class; Y = savings a year later.

(b) Which direction is the selection bias? Why?

> Positive. People who sign up for a savings class are likely already careful with money and would save more anyway: $E[Y(0, U) \mid S = 1] > E[Y(0, U) \mid S = 0]$. The naive comparison overstates the class's effect.

(c) Why would random assignment remove the bias and make ATT = ATE?

> Randomization gives $S \perp U$, so U has the same distribution in both groups. Then $E[Y(0, U) \mid S = 1] = E[Y(0, U) \mid S = 0]$, and conditioning on S = 1 no longer changes the average effect, so ATT = ATE.

Problem 6. Who benefits?

A voluntary job-training program: $S = 1$ for people who enroll, with $P(S = 1) = 0.2$. $Y(S, U)$ is annual earnings. Define

$$ATT = E[Y(1, U) - Y(0, U) \mid S = 1] \quad \text{(effect on those who enroll)}$$
$$ATU = E[Y(1, U) - Y(0, U) \mid S = 0] \quad \text{(effect on those who do not)}$$

(a) Using the LIE, show that $ATE = P(S = 1) \cdot ATT + P(S = 0) \cdot ATU$.

(b) Suppose $ATT = \$3{,}000$ and $ATU = \$500$. Compute the ATE. Why is it so much closer to the ATU?

(c) Officials see an ATT of \$3,000 and expect that making the program mandatory will raise earnings by \$3,000 per person. What is wrong with this reasoning?

Problem 6. Solution (a)

Setup: voluntary job training, $P(S = 1) = 0.2$. ATT = effect on enrollees, ATU = effect on non-enrollees.

(a) Show $ATE = P(S = 1)\cdot ATT + P(S = 0)\cdot ATU$.

> Let $\Delta = Y(1, U) - Y(0, U)$, each person's own effect. So ATE = $E[\Delta]$, ATT = $E[\Delta \mid S = 1]$, ATU = $E[\Delta \mid S = 0]$.
> $ATE = E[\Delta] = E[\, E[\Delta \mid S]\,]$ (LIE)
> $= E[\Delta \mid S = 1]\cdot P(S = 1) + E[\Delta \mid S = 0]\cdot P(S = 0)$ (S is binary: two groups)
> $= P(S = 1)\cdot ATT + P(S = 0)\cdot ATU$

In words: the average effect over everyone is each group's average effect, weighted by group size. This is the practice exam's Springsteen question with enrollees and non-enrollees in place of fans and non-fans.

Problem 6. Solution (b–c)

Setup: ATE = P(S = 1)·ATT + P(S = 0)·ATU, with P(S = 1) = 0.2.

(b) ATT = \$3,000, ATU = \$500. Compute the ATE. Why is it closer to the ATU?

> ATE = 0.2 × 3,000 + 0.8 × 500 = 600 + 400 = \$1,000. The weights are group shares, and 80% of people do not enroll, so their smaller effect dominates the average.

(c) Why won't a mandatory program raise earnings by \$3,000 per person?

> Effects differ across people. Those who chose to enroll likely gain the most. A mandatory program mostly reaches people who would not have enrolled, whose effect is \$500, so the average gain is the ATE, about \$1,000. The right parameter depends on the policy question.

Practice exam link: with 95% fans (positive effect) and a similar-sized negative effect on the other 5%, the ATE is positive, because the fans' term carries 0.95 of the weight.

Problem 7. Where does Cov/Var come from?

The best linear predictor picks β_0 and β_1 to minimize $E[(Y - \beta_0 - \beta_1 X)^2]$. Setting the derivatives to zero gives two first-order conditions:

$$E[Y - \beta_0 - \beta_1 X] = 0$$
$$E[X(Y - \beta_0 - \beta_1 X)] = 0$$

(a) Solve the first condition for β_0.

(b) Substitute into the second condition and show that $\beta_1 = \text{Cov}(X, Y) / \text{Var}(X)$.

(c) Write $\varepsilon = Y - \beta_0 - \beta_1 X$. What do the two conditions say about ε? Do they require any causal assumption?

Problem 7. Solution (a–b)

Setup: first-order conditions $E[Y - \beta_0 - \beta_1 X] = 0$ and $E[X(Y - \beta_0 - \beta_1 X)] = 0$.

(a) Solve the first condition for β_0.

> $E[Y] - \beta_0 - \beta_1 E[X] = 0$ (linearity), so $\beta_0 = E[Y] - \beta_1 E[X]$.

(b) Substitute into the second condition; show $\beta_1 = Cov(X, Y) / Var(X)$.

> $E[XY] - \beta_0 E[X] - \beta_1 E[X^2] = 0$ (linearity)
> $E[XY] - (E[Y] - \beta_1 E[X])E[X] - \beta_1 E[X^2] = 0$ (substitute β_0 from part a)
> $(E[XY] - E[X]E[Y]) - \beta_1 (E[X^2] - E[X]^2) = 0$, i.e. $Cov(X, Y) - \beta_1 Var(X) = 0$
> So $\beta_1 = Cov(X, Y) / Var(X)$.

Problem 7. Solution (c)

(c) What do the conditions say about ε? Any causal assumption needed?

> $E[\varepsilon] = 0$ and $E[X\varepsilon] = 0$: the residual averages zero and is uncorrelated with X. These hold by construction for any data, with no causal assumption. That is why the descriptive slope is always identified, and why it says nothing about whether $S \perp U$.

Link to the practice exam (T/F 5): even when S is not independent of U, we can always estimate the descriptive slope. We just can't call it causal.

Problem 8. Regression with a binary treatment

The causal model is $Y(S, U) = \alpha_0 + \alpha_1 S + U$, where S is binary with $P(S = 1) = p$.

(a) Show that $\mathrm{Cov}(S, Y) = p(1 - p)(E[Y \mid S = 1] - E[Y \mid S = 0])$.

(b) Show that $\mathrm{Var}(S) = p(1 - p)$.

(c) What does the regression slope $\beta_1 = \mathrm{Cov}(S, Y) / \mathrm{Var}(S)$ equal? Under what assumption does it equal the causal effect α_1, the ATE?

Hint: use the LIE on both $E[SY]$ and $E[Y]$, and note that $SY = 0$ whenever $S = 0$.

Problem 8. Solution (a)

(a) Show Cov(S, Y) = p(1 − p)(E[Y | S = 1] − E[Y | S = 0]).

1. $E[SY] = E[SY \mid S = 1]\cdot p + E[SY \mid S = 0]\cdot(1 - p) = E[Y \mid S = 1]\cdot p$ (LIE; SY = 0 when S = 0)
2. $E[S]E[Y] = p\cdot(E[Y \mid S = 1]\cdot p + E[Y \mid S = 0]\cdot(1 - p))$ (LIE on E[Y]) **remember E[S] = P(S = 1) = p when S is binary
3. $\mathrm{Cov}(S, Y) = E[Y \mid S = 1]\cdot p - E[Y \mid S = 1]\cdot p^2 - E[Y \mid S = 0]\cdot p(1 - p)$
$= p(1 - p)(E[Y \mid S = 1] - E[Y \mid S = 0])$

In words: Step 1: split E[SY] into the treated and control groups. In the treated group S = 1, so SY is just Y; in the control group S = 0, so SY = 0 and that group drops out. Step 2: split E[Y] the same way, then multiply by E[S] = p. Step 3: subtract, then factor out p(1 − p), using $p - p^2 = p(1 - p)$.

Problem 8. Solution (b–c)

(b) Show Var(S) = p(1 − p).

> $S^2 = S$ when S is 0 or 1, so $E[S^2] = p$. Then $Var(S) = E[S^2] - E[S]^2 = p - p^2 = p(1 - p)$.

(c) What does β1 equal? When does it equal α1 = ATE?

> $\beta_1 = E[Y \mid S = 1] - E[Y \mid S = 0]$, the difference in group means. That is always a valid descriptive slope. It equals $\alpha_1 =$ ATE only if $S \perp U$, as in an experiment or natural experiment.

Putting it together: $\beta_1 = Cov(S, Y) / Var(S) = p(1 - p)(E[Y \mid S = 1] - E[Y \mid S = 0]) / p(1 - p)$. The $p(1 - p)$ cancels, leaving the difference in means.

Problem 9. Omitted variables

You regress test scores Y on weekly hours of tutoring X using the descriptive model $Y = \beta_0 + \beta_1 X + \varepsilon$. The true causal formula is

$$Y = 2X - Z + \eta$$

where Z is weekly hours of part-time work. Students who work more get less tutoring, so $\mathrm{Cov}(X, Z) < 0$. The noise η satisfies $\eta \perp X$ and $\eta \perp Z$.

(a) Is β_1 larger or smaller than the true causal effect of tutoring, 2? Give a complete mathematical justification.

(b) Under what condition would β_1 equal 2?

Problem 9. Solution

Setup: $Y = 2X - Z + \eta$, $\text{Cov}(X, Z) < 0$, $\eta \perp X$, $\eta \perp Z$. We regress Y on X alone.

(a) Is β_1 larger or smaller than 2? Justify fully.

Step	Justification
$\beta_1 = \text{Cov}(2X - Z + \eta, X) / \text{Var}(X)$	Definition of β_1; substitute Y
$= [2\text{Cov}(X, X) - \text{Cov}(Z, X) + \text{Cov}(\eta, X)] / \text{Var}(X)$	Covariance properties (iii), (iv)
$= [2\text{Var}(X) - \text{Cov}(X, Z) + 0] / \text{Var}(X)$	$\text{Cov}(X, X) = \text{Var}(X)$; $\eta \perp X \Rightarrow \text{Cov} = 0$
$= 2 - \text{Cov}(X, Z) / \text{Var}(X)$	Algebra

> $\text{Cov}(X, Z) < 0$ and $\text{Var}(X) > 0$, so we subtract a negative, and get a value greater than 2: **$\beta_1 > 2$**. Know the intuition: students with more tutoring also work less, and working less raises scores on its own. The slope credits tutoring for both.

(b) When would β_1 equal 2?

> When $\text{Cov}(X, Z) = 0$: the omitted variable is uncorrelated with X.

Problem 10. Standard errors

A charity randomizes 50 schools: 25 get a free breakfast program and 25 do not. Researchers collect test scores for all 5,000 students, regress score on treatment, and report robust (HC0) standard errors. Estimate: +4.1 points, $p = 0.01$.

(a) What is wrong with their standard errors, and how should they fix it?

(b) After the fix, what happens to the point estimate? What do you expect to happen to the p-value?

(c) Suppose instead only 4 schools had been randomized. Why would even clustered standard errors be unreliable?

Problem 10. Solution

Setup: 50 schools randomized, 5,000 students, robust SEs. Estimate +4.1 points, p = 0.01.

(a) What is wrong with the standard errors? How do you fix it?

> Treatment was assigned by school, and students in the same school share teachers, cafeterias and local shocks, so they are not independent. Robust SEs treat them as 5,000 independent observations and overstate precision. Cluster at the level of assignment: the school.

(b) What happens to the point estimate and the p-value?

> The estimate stays exactly 4.1. The standard error typically grows when clustering, so the p-value rises (microfinance: 0.050 became 0.058).

(c) Why are clustered SEs unreliable with only 4 schools?

> Clustered SEs need many independent clusters; the number of clusters acts as the sample size. With 4 schools we effectively have 4 observations, and any difference is confounded with everything else that differs between those schools.

Problem 11. Reading a regression

In the Dutch Postcode Lottery data, we regress durable-goods spending (€) six months later on being in a winning postcode, clustering by postcode. The coefficient is 627 (slope) with a clustered standard error of 295.

(a) Compute the z-statistic and an approximate 95% confidence interval. Interpret the estimate in one sentence.

(b) Do you reject Ho: effect = 0 at the 5% level? At the 1% level?

(c) Suppose the authors also test 20 other spending categories in which the true effect is zero. How many "significant" results at 5% should we expect by chance?

Problem 11. Solution (a)

Setup: durable spending on win_code, clustered by postcode. Coefficient 627, SE 295.

(a) z-statistic, 95% CI and interpretation.

> z = 627 / 295 ≈ 2.13 = (slope-H0)/standard error = (slope - 0)/ SE = slope /SE in this case. CI ≈ 627 ± 1.96 × 295 = 627 ± 578 = slope estimate +/- critical value times SE) roughly [49, 1205]. Being in a winning postcode raises durable spending by about €627 on average six months later. The interval is wide: we are fairly sure the effect is positive, much less sure of its size.

Problem 11. Solution (b–c)

Setup: durable spending on win_code, clustered by postcode. Coefficient 627, SE 295, $z \approx 2.13$.

(b) Reject at 5%? At 1%?

> $|z| = 2.13 > 1.96$, so reject at 5% ($p \approx 0.03$). $|z| < 2.58$, so fail to reject at 1%. You can also find the p-value and then reject / not reject.

(c) 20 more tests with true effects of zero: how many "significant" at 5%?

> 20×0.05 (significance level, type I error tolerance) = 1 false rejection expected by chance. Same logic as a balance table: one or two starred rows out of many are what chance alone produces.

Remember: the p-value is computed assuming the **null** is true, not the alternative. That is why a small p-value is evidence against the null.

Problem 12. A housing lottery

New York City allocates affordable apartments by lottery among people who apply. Researchers compare children of lottery winners and losers to estimate the effect of getting an apartment on school attendance.

(a) Why is winning as-if random among applicants, but not among all NYC families?

(b) Some winners turn down the apartment. Which assumption fails? What parameter can we still identify?

(c) Many winners move into the same new buildings. Describe one way SUTVA could fail.

(d) A balance test shows 1 of 12 pre-lottery characteristics differs at the 5% level. Is this a concern? What can a balance test never rule out?

Problem 12. Solution (a–b)

Setup: NYC affordable-housing lottery among applicants; outcome is children's school attendance.

(a) Why is winning as-if random among applicants, but not among all NYC families?

> Applying is a choice. Applicants may differ from non-applicants in housing need, motivation or information, so $S \perp U$ fails in the whole population. Among applicants, the lottery alone decides who wins, so winning is independent of U. Define the population of interest as applicants, just as Kuhn et al. restrict to lottery players.

(b) Some winners turn down the apartment. Which assumption fails? What can we still identify?

> Perfect compliance fails: winning (assignment) is not the same as moving in (treatment), and who declines is probably related to U. We can still identify the ITT, the effect of winning the lottery, by comparing winners and losers as assigned.

Problem 12. Solution (c–d)

Setup: NYC affordable-housing lottery among applicants; outcome is children's school attendance.

(c) Many winners move into the same new buildings. Describe one way SUTVA could fail.

> One child's outcome could depend on other children's treatment. Winners' children may share a new school and influence each other's attendance, or losing families may lose neighbors, carpools or classmates when winners move out. Fixes: define treatment at a group level, or argue the spillovers are small.

(d) 1 of 12 characteristics differs at 5%. A concern? What can a balance test never rule out?

> Not on its own: with 12 tests at 5%, we expect $12 \times 0.05 = 0.6$ false rejections by chance. But a balance test can never rule out differences in unobservables, such as a rigged lottery. That needs institutional knowledge of how the lottery is run.

How to answer a diagnostic question

Check	Ask	Red flags
$S \perp U$	Was treatment truly random? What does the balance table show?	Self-selection, “first come” or chosen assignment
Compliance	Did everyone take the treatment they were assigned?	Optional take-up, people switching groups
SUTVA	Can one unit’s treatment affect another’s outcome?	Shared classrooms or offices, contagion
Estimation	Is n large enough? Are observations independent?	Tiny samples; clustered data with robust SEs
Interpretation	Does the conclusion follow from the numbers?	“Not significant = no effect”; ITT read as ATE

Pick the **biggest** problem, name the assumption, explain the mechanism and what it does to the estimate, then suggest a fix. Practice exam: **SUTVA**. Phones ring in shared classrooms, so control students’ phones distract treated students; fix by randomizing whole classrooms.

Diagnostic 1. Bike lanes

The scenario has one major flaw. Name it using the language from class and explain why it is a substantial problem.

A city wants to know whether protected bike lanes reduce cyclist injuries. In 2025 it installed lanes on the streets whose community boards submitted requests first, an order the city calls "essentially arbitrary." Researchers treat this as a natural experiment and compare 2026 injury rates on streets with and without new lanes.

Table 1: Balance test

Characteristic	Lanes	No lanes	p
Street length (km)	1.42	1.38	0.71
Speed limit (mph)	25.4	25.9	0.48
2024 traffic (000 cars/day)	11.2	10.8	0.62
Streets	84	312	

Table 2: Injuries per 1,000 cyclists, 2026

Lanes	No lanes	Difference (SE)	p
4.1	5.0	−0.9 (0.3)	0.003

The researchers conclude: "Bike lanes cut injuries by 18%. The balance test confirms treatment was as-if random, so this is the causal effect."

Diagnostic 1. Model answer

Setup: lanes went to streets whose boards requested first; balance on 3 observables; −0.9 injuries per 1,000 (18%), p = 0.003.

Flaw	Lanes went to streets whose boards asked first. Treatment was chosen, not as-if random.
Assumption violated	$S \perp U$, so there is selection bias.
Why it matters	Boards that organize quickly likely have more cycling advocacy, money or past safety work. Those factors sit in U and lower injuries anyway: $E[Y(0, U) \mid S = 1] < E[Y(0, U) \mid S = 0]$.
Effect on the estimate	The −0.9 mixes the effect of lanes with selection bias, and most likely overstates the benefit.
Why Table 1 does not help	It checks only three observables. Advocacy and civic engagement are unobserved, and "essentially arbitrary" is a claim, not a lottery.
What is fine	Compliance and SUTVA are not the issue, and with 396 streets the estimate is reasonably precise.
Fix	Use real randomness, such as a lottery among streets that requested lanes, and compare within requesters.

Diagnostic 2. Flu shots at work

The scenario has one major flaw. Name it using the language from class and explain why it is a substantial problem.

A company wants to know whether free flu shots reduce sick days. Within each office, half of the employees are randomly chosen to receive a free shot; the rest are offered nothing. Everyone chosen gets the shot, and no one else does.

Table 1: Balance test

Characteristic	Treated	Control	p
Age (years)	38.2	38.5	0.74
Manager (=1)	0.21	0.20	0.81
Prior-year sick days	3.6	3.7	0.69
Employees	400	400	

Table 2: Sick days this winter

Treated	Control	Difference (SE)	p
3.1	3.4	−0.3 (0.1)	0.003

The company concludes: “Flu shots reduce sick days by only 0.3 days per employee. A company-wide program is not worth the cost.”

Diagnostic 2. Model answer

Setup: flu shots randomized within offices; good balance and compliance; −0.3 sick days, p = 0.003.

Flaw	A vaccine for a contagious illness is randomized to individuals within the same offices.
Assumption violated	SUTVA: one person's treatment affects another's outcome (spillovers).
Why it matters	Vaccinated coworkers protect the unvaccinated, so a control employee's sick days depend on others' shots. The control group is partly "treated."
Effect on the estimate	Control sick days fall below the true no-vaccine level, so −0.3 understates what a company-wide program would do.
What is fine	Randomization, balance (Table 1) and compliance all look good. The problem is the unit of randomization.
Fix	Randomize whole offices and cluster SEs by office. The estimate then includes direct and spillover effects, which is what a company-wide policy delivers.

Diagnostic 3. A math tutoring app

The scenario has one major flaw. Name it using the language from class and explain why it is a substantial problem.

A school district pilots a math tutoring app. Sixty students are randomly split: 30 get the app for one semester and 30 do not. Every student assigned the app uses it as intended, and no one else has access.

Table 1: Balance test

Characteristic	App	No app	p
Prior math score	71.2	70.8	0.88
Female (=1)	0.53	0.50	0.80
Grade level	10.1	10.0	0.70
Students	30	30	

Table 2: End-of-term math score

App	No app	Difference (SE)	p
76.0	70.0	6.0 (4.6)	0.19

The district concludes: “The difference is not statistically significant, so the app has no effect on math scores. We will not adopt it.”

Diagnostic 3. Model answer

Setup: 60 students randomized; good balance and compliance; +6.0 points (SE 4.6), $p = 0.19$.

Flaw	Treating “not statistically significant” as “no effect.”
Concept	Misinterpreting a hypothesis test. With so few students the test has low power, so a Type II error is likely.
Evidence in Table 2	The 95% CI is about $6.0 \pm 1.96 \times 4.6 = 6.0 \pm 9.0$, roughly $[-3, 15]$. It includes zero, but also large positive effects.
Why it matters	Failing to reject Ho is not the same as showing Ho is true. The data cannot tell a zero effect apart from a 15-point gain.
What is fine	Randomization (good balance), compliance and SUTVA all look fine. The design is valid; the reading of the result is not.
What they can say	The estimate is positive but too imprecise to conclude either way.
Fix	Run a larger trial before deciding, so the standard error is small enough to be informative.

Summary: what each design needs

Setting	What you need	How to check	What goes wrong
Randomized experiment	$S \perp U$, SUTVA, perfect compliance	Balance test on observables	Non-compliance: estimate the ITT. Spillovers: cluster-randomize
Natural experiment	Treatment as-if random within a well-defined population	Balance test plus institutional context	Self-selection into the pool; unobservables a balance test cannot see
Causal regression	$S \perp U$ ($E[SU] = 0$)	Ask what is hiding in U	Omitted variable bias
Inference	Independent observations, or many independent clusters	At what level was treatment assigned?	Robust SEs overstate precision under clustered assignment

On the diagnostic: pick one flaw, name the assumption, and explain what it does to the estimate.